

Numerical PDE Quiz 1

March 18, 2026

Consider the differential equation

$$-u''(x) + u(x) = f(x), \quad x \in (0, 1),$$

with boundary conditions $u(0) = u(1) = 0$.

1. Sketch the basic procedures to solve the differential equation by the finite difference method.
2. Find all the eigenpairs of the matrix A formed in last sub-question.
3. Show the L^2 -stability
4. If $f(x) > 0$, show that $u_j > 0$.